ModelIt!

Teacher's Guide

# Why Do Magnets Stick?

Exploring Magnetic Forces and Fields

## Quick Reference

| | |
|---|---|
| **Grade Level** | Grade 3 |
| **NGSS Standard** | 3-PS2-3 |
| **Time** | 45–60 minutes (core lesson) |
| **Materials** | Student Activity Pack, devices with ModelIt! access |
| **Prep** | Print activity packs, test ModelIt! on student devices |
| **Grouping** | Partners or small groups (2–4) |

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# 1 NGSS Standards & CAST Alignment

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**Performance Expectation: 3-PS2-3**

Ask questions to determine cause and effect relationships of electric or magnetic interactions between two objects not in contact with each other.

## Three-Dimensional Learning

| | |
|---|---|
| **Science Practice** | **Asking Questions and Defining Problems** — Students ask questions about why magnets interact with some materials and not others, and investigate cause and effect relationships in magnetic systems. |
| **Disciplinary Core Idea** | **PS2.B Types of Interactions** — Electric and magnetic forces between a pair of objects do not require that the objects be in contact. The sizes of the forces in each situation depend on the properties of the objects and their distances apart. |
| **Crosscutting Concept** | **Cause and Effect** — Students investigate how changing magnet strength (cause) affects magnetic force and attraction (effects) in a predictable pattern. |

## CAST Testing Alignment

**MULTIPLE CHOICE:**

A student holds a magnet near a paperclip, a wooden block, and a plastic cup. Only the paperclip moves toward the magnet. Which best explains why?

**CONSTRUCTED RESPONSE:**

Using what you know about magnetic force, explain why a strong magnet can pick up a paperclip from farther away than a weak magnet can. Use the words 'magnet strength' and 'magnetic force' in your answer.

# 2 Scoring Rubric

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### Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

### Scoring Guidance

**Expert (4):** Student demonstrates deep understanding and can transfer knowledge to new contexts.
**Proficient (3):** Student meets grade-level expectations and shows solid understanding.
**Developing (2):** Student is making progress but needs additional scaffolding or practice.
**Beginning (1):** Student needs significant support; consider using Differentiation strategies (Section 7).

# 3 Background Content Knowledge

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**For Your Confidence**

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Magnetism is one of nature's most fascinating forces. It is invisible, acts at a distance, and has been used by humans for thousands of years — from ancient compasses to modern electric motors and MRI machines in hospitals.

## What Is a Magnet?

A magnet is an object that produces an invisible magnetic field around itself. This field can attract (pull toward) or repel (push away) other magnets and certain metals. Every magnet has two poles — a north pole and a south pole. Opposite poles attract each other, while same poles repel. The magnetic field is strongest at the poles.

## Why Do Magnets Attract Some Things?

Magnets only attract metals that contain iron, nickel, or cobalt. Inside these metals, tiny groups of atoms called magnetic domains can line up in the same direction when a magnet is nearby. When the domains line up, the metal becomes temporarily magnetic itself and is pulled toward the magnet. Materials like wood, plastic, and glass do not have magnetic domains, so magnets have no effect on them.

## Magnetic Force and Distance

Magnetic force follows an important rule: the closer the magnet is to the object, the stronger the pull. As the distance doubles, the force drops dramatically. A strong magnet might pick up a paperclip from two inches away, but at four inches, the force may be too weak. This is why you have to get a fridge magnet close to the fridge for it to stick — the force needs to be strong enough to hold it up against gravity.

# 4 The LEVER Framework

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### About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

**L** **Locate the System**
Students identify magnet strength as an external component and magnetic force and attraction as internal components that change in response to how powerful the magnet is.

**E** **Establish Relationships**
Students determine that magnet strength positively affects magnetic force, and magnetic force positively affects attraction — creating a direct chain of cause and effect.

**V** **Visualize & Model**
Students build a model showing how increasing magnet strength leads to greater magnetic force and stronger attraction to metal objects.

**E** **Evaluate Outcomes**
Students run weak magnet, medium magnet, and strong magnet scenarios to observe how changing one input (strength) affects the entire system.

**R** **Revise & Extend**
Students add distance or object material to explore why magnets work better up close and only on certain metals.

## Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

| | |
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| **At the start of a unit** | Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it. |
| **After a reading or lab** | Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes. |
| **As a "what-if" tool** | When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back. |
| **For peer discussion** | Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence. |
| **As a presentation tool** | Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time. |
| **At the end of a unit** | Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved. |

# 5 Slide-by-Slide Facilitation

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## Slide 1: Cover

2 min

**Say:** *"Who has ever put a magnet on the refrigerator? Did you ever wonder HOW it sticks up there without falling?"*

**Watch For**

Pass out magnets and let students test them on their desks, pencils, and zippers for 30 seconds.

## Slide 2: What Will We Learn Today?

3 min

**Say:** *"Today we are going to figure out the mystery of magnets — why they stick to some things but not others!"*

**Watch For**

Read objectives together. Introduce vocabulary with a quick demo: hold a magnet near a paperclip and ask what force is acting.

## Slide 3: The Big Question

3 min

**Say:** *"Here is our big question: why does a magnet grab a paperclip but totally ignore a pencil?"*

**Watch For**

Have students turn and talk with a partner: What do YOU think makes a magnet stick to some things? Collect ideas on the board.

## Slide 4: Let's Build a Model!

3 min

**Say:** *"We are going to build a model to solve this magnet mystery — step by step, like real scientists!"*

**Watch For**

Preview each LEVER step briefly. Remind students that a model helps us understand how things work.

## Slide 5: Activity 1: Sort the Parts

7 min

**Say:** *"What parts of the magnet system can we identify? What can we control, and what happens on its own?"*

**Watch For**

Guide sorting of three components. Explain that magnet strength is external because WE choose which magnet to use.

## Slide 6: Activity 2: Connect with Arrows

7 min

**Say:** *"When we use a stronger magnet, does the pulling force go up or down? Let us draw our arrows!"*

**Watch For**

Help students draw positive arrows from magnet strength to magnetic force, and from magnetic force to attraction.

## Slide 7: Activity 3: Run the Simulation!

8 min

**Say:** *"Time to test our model! Let us start with a weak magnet and see what happens."*

**Watch For**

Have students predict first, then run each scenario. Compare weak vs. medium vs. strong magnet results on the graph.

## Slide 8: What Did We Discover?

7 min

**Say:** *"Now we know the secret — magnets have an invisible force that only works on certain metals!"*

**Watch For**

Summarize findings. Demo: sprinkle iron filings around a magnet to make the invisible field visible. Students draw what they see.

# 6 Answer Key & Expected Responses

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## Component Sorting

- **EXTERNAL:** Magnet Strength
- **INTERNAL:** Magnetic Force, Attraction

## Relationships

- **Magnet Strength to Magnetic Force:** POSITIVE (+) — A stronger magnet creates a bigger invisible field with more pulling power. When you switch from a weak magnet to a strong one, the magnetic force increases — you can feel the pull get stronger.
- **Magnetic Force to Attraction:** POSITIVE (+) — When the magnetic force is stronger, the magnet pulls harder on metal objects. More force means stronger attraction — the paperclip jumps to the magnet from farther away.

## Key Discoveries

- Stronger magnets create a stronger magnetic force that can pull from farther away
- Magnets only attract certain metals like iron, nickel, and cobalt — not plastic, wood, or paper
- Magnetic force can work through air without the magnet touching the object
- The closer a magnet gets to a metal object, the stronger the pull feels

# 7 Collaborative Learning Protocols

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### Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

### Model Gallery Walk

**When:** After students build models (LEVER Phase V)
**Setup:** Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
**Protocol:** Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
**Why it works:** Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

### Scientist Fishbowl

**When:** During simulation discussion (LEVER Phase E)
**Setup:** 4-5 students sit in center circle. Rest of class observes with clipboards.
**Protocol:** Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
**Why it works:** Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

### Component Jigsaw

**When:** During component analysis (LEVER Phase L)
**Setup:** Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
**Protocol:** Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
**Why it works:** Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

### Prediction Debate

**When:** Before running simulations (LEVER Phase E)
**Setup:** Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
**Protocol:** Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
**Why it works:** Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

### Pair Programming for Models (Digital Collaboration)

**When:** Throughout model building
**Setup:** Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
**Protocol:** Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
**Why it works:** Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

### Quick Protocol Selection Guide

| | |
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| **Short on time (5 min)** | Prediction Debate — gets bodies moving, generates energy fast |
| **Full period available** | Component Jigsaw — deep collaboration, every student contributes |

| | |
|---|---|
| **After simulation day** | Model Gallery Walk — peer feedback, multiple perspectives |
| **Advanced discussion** | Scientist Fishbowl — builds academic discourse skills |
| **Every day default** | Pair Programming — transforms individual work into dialogue |

# 8 Differentiation & Extensions

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### Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

### Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

### English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

### Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

# 9 Common Misconceptions

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### Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

| STUDENTS MAY THINK | SCIENTIFIC UNDERSTANDING |
|---|---|
| "Magnets stick to all metals" | Magnets only attract metals that contain iron, nickel, or cobalt. Many metals like aluminum (soda cans), copper (pennies), and gold (jewelry) are NOT magnetic at all. Test different metals to see which ones a magnet attracts. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "Magnets lose their power over time just from using them" | Permanent magnets keep their magnetism for a very long time. They can lose strength if they are dropped hard, heated up, or stored incorrectly, but normal use does not weaken them. The magnetic field is created by the arrangement of atoms inside the magnet. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "Bigger magnets are always stronger" | Size does not always determine magnet strength. A small neodymium magnet can be much stronger than a large ceramic magnet. What matters is the material the magnet is made from and how its atoms are arranged inside, not just how big it is. |